Phillip Lippe

PhD student fascinated by research in artificial intelligence and deep learning, especially causality and representation learning. Interested in exploring new research areas and connecting with other researchers.

Gender **Male** Age **24 years** Nationality **German** Languages **German (native), English (C1)** **Amsterdam, Netherlands**

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EXPERIENCE

Student Research Assistant

University of Amsterdam / Ahold Delhaize

Jul 2019 – Dec 2019 Amsterdam, Netherlands

- Research in task-oriented dialogue system at the ILPS and AI Retail lab in collaboration with Ahold Delhaize
- Research topic: Diversifying Dialogue Response Generation with Prototype Guided Paraphrasing (publication)

Student Research Assistant

Vrije Universiteit Amsterdam

Nov 2018 – June 2019 Amsterdam, Netherlands

- Research in theorem proving for high-order logic
- Developing a hammer for translating high-order Lean expressions into first-order logic (project page)

Student Research Intern

Daimler AG, Mercedes-Benz

Dec 2016 – Mar 2017
Apr 2018 – Sep 2018 Stuttgart, Germany

- Research in deep learning for real-time multi-class object detection in complex urban traffic scenes (autonomous driving)
- Bachelor Thesis: *Hierarchical Multi-label Object Detection of Rare Classes for Autonomous Driving* (pdf)

Student Research Intern

Mercedes-Benz Research and Development NA

May 2017 – Aug 2017 Sunnyvale, United States

- Research in deep generative models for predicting agent behavior in traffic scenarios for autonomous driving (report)
- Performed agile software development with Scrum

PROJECTS

Google Developer Expert, Machine Learning

- Member of the ML GDE program for expertise and public content sharing in JAX+Flax since August 2022

Tutorials

- Created and taught a series of implementation tutorials in PyTorch and JAX for various Deep Learning topics (website, ~40k page visits per month, >1k GitHub stars). Part of the DL courses at UvA and official tutorials of PyTorch Lightning

Teaching assistant

- Teaching assistant for the graduate courses Deep Learning, NLP 1, FAIR, IR 1, and Advanced Topics in Computational Semantics at the University of Amsterdam (2019-2022)
- Head Teaching Assistant for ASCI PhD Course on Computer Vision by Learning 2022 (link)

Summer schools

- Participated in the Oxford ML Summer School 2021, and the CIFAR DL and Reinforcement Learning Summer School 2021

EDUCATION

PhD, Artificial Intelligence

University of Amsterdam, QUVA lab

Sep 2020 – (exp.) Sep 2024 Amsterdam, Netherlands

- Supervisors: dr. Efstratios Gavves and dr. Taco Cohen
- My research focuses on the intersection of causality and deep learning, in particular causal representation learning (abstract)
- ELLIS PhD; collaboration with Qualcomm AI Research

Master of Science, Artificial Intelligence

University of Amsterdam

Sep 2018 – Aug 2020 Amsterdam, Netherlands

- Courses on AI, including ML, DL, NLP, IR, CV, and RL
- Thesis on "Categorical Normalizing Flows" (publication), applied for permutation-invariant graph/molecule generation
- Final GPA: 9.5, cum laude (Dutch grading system)

Bachelor of Engineering, Computer Science

Baden-Wuerttemberg Cooperative State University

Oct 2015 – Sep 2018 Stuttgart, Germany

- Cooperative study program with the specialization in IT Automotive and autonomous driving, in cooperation with Daimler AG/Mercedes-Benz R&D
- Final GPA: 1.0 (German grading system)

Pre-Studies, Mathematics/Computer Science

Ruhr University Bochum

Apr 2012 – Sep 2014 Bochum, Germany

- Completing university courses as high school student, including Introduction to Software Engineering I & II, Linear Optimization, and Programming for Mathematicians

AWARDS

- Awarded SchülerUni Scholarship 2012/13 for taking university courses on Computer Science/Mathematics during high school
- Best CS undergraduate student at the Baden-Wuerttemberg Cooperative State University 2018
- Won 4th place in the NeurIPS 2020 challenge "Hateful Memes" of Facebook AI (paper)

SKILLS

DL Frameworks: PyTorch (2018-present, R&T), JAX/Flax (2021-present, R&T), Tensorflow (2017-2018, R), Caffe (2016-2017, R) (R - Research, T - Teaching)

Programming languages: Python (2016-present), HTML/PHP/SQL (2012-present), Java (2012-2019), MATLAB (2016-2019), C (2015-2017)

Additional skills: SLURM cluster computing (2018-present), git (2015-present), Docker (2017-present)

PUBLICATIONS

Conferences

- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **Causal Representation Learning for Instantaneous and Temporal Effects in Interactive Systems**. International Conference on Learning Representations (ICLR), 2023 (link).
- Adeel Pervez, Phillip Lippe, Efstratios Gavves: **Differentiable Mathematical Programming for Object-Centric Representation Learning**. International Conference on Learning Representations (ICLR), 2023 (link).
  — *Own contributions:* Proposed application to object-centric learning. Advised in model development and experimental setups. Contributed to writing
- Adeel Pervez, Phillip Lippe, Efstratios Gavves: **Scalable Subset Sampling with Neural Conditional Poisson Networks**. International Conference on Learning Representations (ICLR), 2023 (link).
  — *Own contributions:* Advised in model development and experimental setups. Contributed to writing
- Johann Brehmer, Pim de Haan, Phillip Lippe, Taco Cohen: **Weakly supervised causal representation learning**. Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022 (link).
  — *Own contributions:* Advised in theory and model development, created datasets, implemented causal discovery pipeline, and contributed to writing
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **CITRIS: Causal Identifiability from Temporal Intervened Sequences**. International Conference on Machine Learning (ICML), 2022 (link).
- Anna Langedijk, Verna Dankers, Phillip Lippe, Sander Bos, Bryan Cardenas Guevara, Helen Yannakoudakis, and Ekaterina Shutova: **Meta-learning for fast cross-lingual adaptation in dependency parsing**. Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL, Volume 1: Long Papers), 2022 (link).
  — *Own contributions:* Advised in training and finetuning of large language Transformers, guided as teaching assistant through the project
- Phillip Lippe, Taco Cohen and Efstratios Gavves: **Efficient Neural Causal Discovery without Acyclicity Constraints**. International Conference on Learning Representations (ICLR), 2022 (link).
- Phillip Lippe and Efstratios Gavves: **Categorical Normalizing Flows via Continuous Transformations**. International Conference on Learning Representations (ICLR), 2021 (link).

Journals

- Sindy Löwe, Phillip Lippe, Maja Rudolph, Max Welling: **Complex-Valued Autoencoders for Object Discovery**. Transactions on Machine Learning Research, Nov 2022 (link).
  — *Own contributions:* Advised in model development, experiment design and model optimization. Contributed to writing
- Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Casey A. Fitzpatrick, Peter Bull, Greg Lipstein, Tony Nelli, Ron Zhu, Niklas Muennighoff, Rize Velioglu, Jewgeni Rose, Phillip Lippe, Nithin Holla, Shantanu Chandra, Santhosh Rajamanickam, Georgios Antoniou, Ekaterina Shutova, Helen Yannakoudakis, Vlad Sandulescu, Umut Ozertem, Patrick Pantel, Lucia Specia, Devi Parikh (2021). **The Hateful Memes Challenge: Competition Report.** Proceedings of the NeurIPS 2020 Competition and Demonstration Track, PMLR 133:344-360, 2021 (link).
  — *Own contributions:* Added summary of own solution and results to the Hateful Memes Challenge

Workshops

- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **Intervention Design for Causal Representation learning**. First Workshop on Causal Representation Learning (CRL), UAI 2022 (link).
- Phillip Lippe, Pengjie Ren, Hinda Haned, Bart Voorn and Maarten de Rijke: **Simultaneously Improving Utility and User Experience in Task-oriented Dialogue Systems**. SIGIR Workshop on eCommerce (eCom), 2022 (link). *Contributed talk*
- Julian Suk, Pim de Haan, Phillip Lippe, Christoph Brune and Jelmer M. Wolterink: **Mesh convolutional neural networks for wall shear stress estimation in 3D artery models**. MICCAI, Workshop on Statistical Atlases and Computational Modelling of the Heart (STACOM), and Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS), 2021 (link).
  — *Own contributions:* Implemented and conducted experiments on point-cloud based baselines, advised for model optimization on other experiments
- Phillip Lippe, Nithin Holla, Shantanu Chandra, Santhosh Rajamanickam, Georgios Antoniou, Ekaterina Shutova, and Helen Yannakoudakis: **A Multimodal Framework for the Detection of Hateful Memes**. NeurIPS 2020 Competition Track, Facebook AI Hateful Memes (link).

PUBLIC TALKS

02/2023 Invited Talk at the Rising Stars in AI Symposium at KAUST on "Causal Representation Learning"
12/2022 Invited Talk at the Google Student Developer Club, University of Augsburg on "Machine Learning with JAX and Flax"
10/2022 Invited Talk at the Innovation Center for Artificial Intelligence on "Causal Representation Learning"
08/2022 Invited Talk at the First Workshop on Causal Representation Learning at UAI 2022 on "Learning Causal Variables from Temporal Sequences with Interventions"
07/2022 Spotlight Talk at ICML 2022 on our paper CITRIS: Causal Identifiability from Temporal Intervened Sequences
07/2022 Contributed Talk at the Workshop On eCommerce at SIGIR 2022
09/2021 Invited Talk at the Innovation Center for Artificial Intelligence on "Neural Causal Discovery"
07/2021 Contributed Talk at the 8th Causal Inference Workshop at UAI 2021
12/2020 Contributed Talk at the NeurIPS 2020 Hateful Memes Competition workshop on our 4th place winning solution
02/2020 Invited Talk at the Innovation Center for Artificial Intelligence on "Diverse Dialogue Response Generation"

REVIEWING

I have served as reviewer for the following conferences and workshops: ECCV-2020, ICCV-2021, CausalUAI-2021, ICLR-2022, CVPR-2022, CLeaR-2022, NeurIPS-2022, CRL-2022, CML4Impact-2022, CDS-2022, nCSI-2022, ICLR-2023, CLeaR-2023